Algebraic and Geometric Reconstruction:
A Bridge via Conditional Variance

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Abstract

Paper IV introduces two witnesses for finite-sample reconstruction: the algebraic witness $\delta(L)$ (worst-case σ-algebra approximation error) and the geometric witness $(\mu\otimes\mu)(R_L)$ (mass of unseparated pairs under the lag-L equivalence relation). This note identifies the exact object mediating between them — the *conditional variance of indicators* — and establishes a two-sided comparison theorem. The easy direction requires no conditions. The hard direction requires that large fibres of the delay map be internally mixed, a condition implied by ergodicity but strictly weaker. As a corollary, reconstruction is equivalent to divergence of the collision entropy of the delay-vector distribution, giving an information-theoretic characterisation of reconstruction not present in Papers I–IV.

1 Setup

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. For $L \geq 0$ define the *observable σ-algebra at lag L*:

$$\mathcal{O}_h^{(L)} \;:=\; \sigma\big(h \circ T^0,\, h \circ T^1,\, \ldots,\, h \circ T^L\big)$$

and the *lag-L delay map* $\Phi_h^{(L)} : X \to \mathbb{R}^{L+1}$, $\Phi_h^{(L)}(x) = (h(x), h(Tx), \ldots, h(T^L x))$.

Definition 1.1 (Algebraic witness)**.** The *σ-algebra approximation error at lag L* is

$$\delta(L) \;:=\; \sup_{S\in\mathcal{B}} \inf_{E\in\mathcal{O}_h^{(L)}} \mu(S \triangle E).$$

Definition 1.2 (Equivalence relation and geometric witness)**.** Define the lag-L equivalence relation $x \sim_L x'$ iff $\Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')$, and the corresponding unseparated-pairs set:

$$R_L \;:=\; \big\{(x, x') \in X \times X : x \sim_L x'\big\}.$$

The *geometric witness* is $(\mu \otimes \mu)(R_L)$.

Definition 1.3 (Fibre structure)**.** For $z \in \mathbb{R}^{L+1}$, let $F_z := (\Phi_h^{(L)})^{-1}(z)$ be the fibre over z, $p_z := \mu(F_z)$ the fibre mass, and μ_z the conditional measure on F_z. The push-forward $\nu_L := (\Phi_h^{(L)})_* \mu$ is the distribution of delay vectors.

Three facts are immediate from the definitions:

(a) $R_L = \bigsqcup_z F_z \times F_z$, so $(\mu \otimes \mu)(R_L) = \int p_z^2 \, d\nu_L(z) = \sum_z p_z^2$ (in the discrete-fibre case).

(b) The conditional variance identity: for any $S \in \mathcal{B}$,

$$\mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] \;=\; \int p_z \cdot \mu_z(S)\mu_z(S^c)\, d\nu_L(z).$$

(c) $\delta(L) = \sup_S \mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})]$, since $\mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] = \|\mathbf{1}_S - \mathbb{E}[\mathbf{1}_S \mid \mathcal{O}_h^{(L)}]\|^2_{L^2(\mu)}$ and $\delta(L) = \sup_S \|\mathbf{1}_S - \mathbb{E}[\mathbf{1}_S \mid \mathcal{O}_h^{(L)}]\|^2_{L^2(\mu)}$ (Paper IV, Lemma 3.1).

2 The Bridge Identity

Lemma 2.1 (Conditional variance identity)**.** *For any $S \in \mathcal{B}$:*

$$\mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] \;=\; \frac{1}{2}\int_{R_L} |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 \, d(\mu\otimes\mu)(x,x').$$

Proof. By disintegration, $(\mu\otimes\mu)|_{R_L} = \int \mu_z \otimes \mu_z \, d\nu_L(z)$. On each fibre F_z:

$$\frac{1}{2}\int_{F_z\times F_z} |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 \, d(\mu_z\otimes\mu_z) \;=\; \mu_z(S)\mu_z(S^c).$$

(Expand: $\int\int |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 \;=\; 2[\mu_z(S)(1-\mu_z(S))]$.) Integrating over fibres with weight $p_z = \mu(F_z)$:

$$\frac{1}{2}\int_{R_L} |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 \, d(\mu\otimes\mu) \;=\; \int p_z \cdot \mu_z(S)\mu_z(S^c)\, d\nu_L(z) \;=\; \mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})]. \qquad \square$$

Remark 2.2. Lemma 2.1 makes precise why the conditional variance is the right mediating object: its left-hand side is the quantity appearing in $\delta(L)$ (algebraic), and its right-hand side is an S-weighted version of $(\mu\otimes\mu)(R_L)$ (geometric). Both measure the same failure of separation, but with different aggregation: $\delta(L)$ takes the worst case over S; $(\mu\otimes\mu)(R_L)$ counts unseparated pairs with equal weight.

3 The Two-Sided Comparison

Theorem 3.1 (Easy direction — unconditional)**.**

$$\delta(L) \;\le\; \tfrac{1}{2}\,(\mu\otimes\mu)(R_L).$$

Proof. From Lemma 2.1 and $|\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 \le 1$:

$$\mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] \;\le\; \tfrac{1}{2}\,(\mu\otimes\mu)(R_L)$$

for every S. Taking the supremum over S gives $\delta(L) \le \frac{1}{2}(\mu\otimes\mu)(R_L)$. $\square$

Remark 3.2. This says: large unseparated-pair mass forces large worst-case error. No dynamical conditions are needed. The constant $\frac{1}{2}$ is sharp: take μ uniform on two points, $T = \mathrm{id}$, h constant. Then $R_L = X\times X$, $(\mu\otimes\mu)(R_L) = 1$, and $\delta(L) = \frac{1}{2}$.

For the reverse direction, the identity in Lemma 2.1 shows that the gap between $(\mu\otimes\mu)(R_L) = \int p_z^2\, d\nu_L(z)$ and $\delta(L) = \sup_S \int p_z\mu_z(S)\mu_z(S^c)\, d\nu_L(z)$ is controlled by whether large fibres (p_z large) are internally mixed.

Definition 3.3 (Fibre mixing condition)**.** Fix $\varepsilon > 0$. Call z an *ε-large fibre* if $p_z \ge \varepsilon$. Say that $S^* \in \mathcal{B}$ is *(c,ε)-mixing across fibres* if there exists $c > 0$ such that

$$\mu_z(S^*)\mu_z((S^*)^c) \;\ge\; c\cdot p_z \quad \text{for } \nu_L\text{-a.e. } \varepsilon\text{-large fibre } z.$$

(The condition is vacuous on fibres with $p_z < \varepsilon$, which contribute at most ε to $(\mu\otimes\mu)(R_L)$.)

Theorem 3.4 (Hard direction — conditional)**.** *Let $\varepsilon > 0$ and suppose S^* is an ε-maximizer of $\delta(L)$, i.e. $\mathbb{E}[\mathrm{Var}(\mathbf{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \ge \delta(L) - \varepsilon$. If S^* is (c,ε)-mixing across fibres, then*

$$(\mu\otimes\mu)(R_L) \;\le\; \frac{1}{c}\,\delta(L) + C(c)\,\varepsilon, \qquad C(c) := \frac{1+c}{c}.$$

In particular, if S^* achieves $\delta(L)$ exactly and is $(c,0)$-mixing, then $(\mu\otimes\mu)(R_L) \le \frac{1}{c}\,\delta(L)$.

Proof. Split $(\mu\otimes\mu)(R_L)=\int p_z^2\,d\nu_L(z)$ into ε-large and small-fibre parts. *Large fibres ($p_z\ge\varepsilon$).* The (c,ε)-mixing condition gives $\mu_z(S^*)\mu_z((S^*)^c)\ge c\,p_z$, so:

$$c\int_{\{p_z\ge\varepsilon\}} p_z^2\,d\nu_L(z) \;\le\; \int p_z\,\mu_z(S^*)\mu_z((S^*)^c)\,d\nu_L(z) \;=\; \mathbb{E}[\mathrm{Var}(\mathbf{1}_{S^*}\mid\mathcal{O}_h^{(L)})] \;\le\; \delta(L).$$

Hence $\int_{\{p_z\ge\varepsilon\}} p_z^2\,d\nu_L(z)\le\frac{1}{c}\,\delta(L)$. *Small fibres ($p_z<\varepsilon$).* Each such fibre satisfies $p_z^2\le\varepsilon\,p_z$, so

$$\int_{\{p_z<\varepsilon\}} p_z^2\,d\nu_L(z) \;\le\; \varepsilon\int p_z\,d\nu_L(z)=\varepsilon.$$

Combining. Since $C(c)=\frac{1+c}{c}\ge 1$, we have $\varepsilon\le C(c)\,\varepsilon$, and therefore:

$$(\mu\otimes\mu)(R_L) \;\le\; \frac{1}{c}\,\delta(L)+C(c)\,\varepsilon.$$

□

Remark 3.5 (Fibre mixing vs ergodicity)**.** The fibre mixing condition (Definition 3.3) is strictly weaker than ergodicity. It requires only that some near-maximizing event S^* straddles large fibres in a non-trivial fraction, not that the whole system is indecomposable. Ergodicity provides a sufficient condition: under an ergodic T, any S^* with $\mathbb{E}[\mathrm{Var}(\mathbf{1}_{S^*}\mid\mathcal{O}_h^{(L)})]>0$ cannot be $\mathcal{O}_h^{(L)}$-measurable, so it must straddle fibres, and ergodicity ensures this holds for any nontrivial near-maximizing sequence. The condition fails only when S^* is aligned with the fibre structure — large fibres entirely inside or entirely outside S^* — an "adversarially structured" failure that ergodicity rules out but strict measure-theoretic conditions do not require to be impossible.

Corollary 3.6 (Equivalence under fibre mixing)**.** *Under the fibre mixing condition:*

$$c\,(\mu\otimes\mu)(R_L) \;\le\; \delta(L) \;\le\; \tfrac{1}{2}\,(\mu\otimes\mu)(R_L).$$

In particular:

$$\delta(L)\to 0 \qquad\Longleftrightarrow\qquad (\mu\otimes\mu)(R_L)\to 0.$$

Proof. Theorem 3.1 gives $\delta(L)\le\frac{1}{2}(\mu\otimes\mu)(R_L)$. For the lower bound, apply Theorem 3.4 with $\varepsilon\to 0$ (taking a sequence of ε_k-maximizers with $\varepsilon_k\to 0$), yielding $(\mu\otimes\mu)(R_L)\le\frac{1}{c}\,\delta(L)$. □

4 Entropy Characterisation of Reconstruction

The geometric witness $(\mu\otimes\mu)(R_L)=\int p_z^2\,d\nu_L(z)$ is the collision probability of the delay-vector distribution ν_L. Its logarithm is the (negative) *Rényi-2 entropy* (collision entropy) of ν_L:

$$H_2(\nu_L) \;:=\; -\log\big((\mu\otimes\mu)(R_L)\big) \;=\; -\log\int p_z^2\,d\nu_L(z).$$

Corollary 4.1 (Entropy characterisation of reconstruction)**.** *Under the fibre mixing condition of Definition 3.3:*

$$\delta(L)\to 0 \qquad\Longleftrightarrow\qquad H_2(\nu_L)\to+\infty.$$

That is, reconstruction holds if and only if the collision entropy of the delay-vector distribution diverges.

Proof. $\delta(L)\to 0 \iff (\mu\otimes\mu)(R_L)\to 0$ (Corollary 3.6) $\iff e^{-H_2(\nu_L)}\to 0 \iff H_2(\nu_L)\to +\infty$. □

Remark 4.2 (Interpretation)**.** Corollary 4.1 gives a purely information-theoretic characterisation of reconstruction: the observable h reconstructs the system $(X,\mathcal{B},\mu,T)$ at lag L if and only if the collision probability of the delay-vector distribution ν_L decays to zero as L grows — equivalently, if and only if the probability that two independently drawn trajectories produce identical delay vectors vanishes.

This characterisation is not present in Papers I–IV. It does not require:

- a metric on X,
- smoothness of T or h,
- a mixing rate,
- or the Takens embedding conditions.

The collision entropy $H_2(\nu_L)$ is estimable from data: it is the negative log of $\sum_z$(fraction of sample in fibre $z)^2$, a standard diversity index computable from the delay-vector histogram. Under Paper IV's standing hypotheses (compact smooth X, smooth positive μ, $T \in C^r$), the fibre mixing condition holds under ergodicity, and the entropy characterisation then provides a third, data-driven witness for reconstruction complementing $\delta(L)$ and the delay-map separation.

Remark 4.3 (Why the empirical algebraic witness $\hat{\delta}(L, n)$ works)**.** Paper IV (§4–5) constructs an empirical witness $\hat{\delta}(L, n)$ and proves that it concentrates around $\delta(L)$ at rate $n^{-1/2}$ (up to logarithms). The conditional variance identity (Lemma 2.1) explains why this is the right quantity to estimate: $\delta(L)$ is the supremum over S of $\mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})]$, and the empirical version replaces the conditional variance with its sample analogue (fibre-conditional fractions computed from the observed delay-vector histogram). Corollary 4.1 now says that the elbow stopping rule $\hat{L}^*$ defined in Paper IV (Thm 5.4) could equivalently be stated as the first L at which the empirical collision entropy $\hat{H}_2(\nu_L^{(n)})$ exceeds a threshold calibrated to n: both witnesses detect the same transition.

Remark 4.4 (Relation to Paper IV witnesses)**.** The three witnesses now in scope are:

(i) $\hat{\delta}(L, n)$ — the algebraic witness (Paper IV, §5);

(ii) $\hat{d}_L(x, x')$ — the delay-map separation (Paper IV, §6);

(iii) $\hat{H}_2(\nu_L^{(n)}) := -\log\big(\frac{1}{n^2}\sum_{i,j}\mathbf{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)]\big)$ — the empirical collision entropy.

All three are computable from the time series alone. Corollaries 3.6 and 4.1 show that (i) and (iii) converge to zero (resp. diverge) together under fibre mixing, making them asymptotically equivalent witnesses for reconstruction. Witness (ii) provides the geometric certificate that the state space is being faithfully separated.

Remark 4.5 (A separation-system direction)**.** The fibre structure studied here suggests a more primitive starting point. Define a *separation system* as a family of equivalence relations $\{\sim_L\}_{L\geq 0}$ on a set X together with a compatible family of charges μ_L on the quotients $X/{\sim_L}$. Reconstruction in this language is the condition $\bigcap_L [x]_L = \{x\}$ for μ-a.e. x. The fibre mixing condition (Definition 3.3) is then a condition on the charge, not on the dynamics: it says the charges μ_L do not concentrate on large equivalence classes. Collective Exhaustion from Paper I (the necessary and sufficient condition for the compatible family to extend to a global measure on X) appears in this language as the condition that the charges μ_L are consistent across levels — a charge-coherence condition, not a dynamical one. Whether this separation-system + coherent-charges framework can serve as a foundation for Papers I–IV, recovering probability and dynamics from primitive discriminability, is a question the present programme leaves open.